

SkateGuard Project Progress Report

Weeks 1 and 2

1. Report Scope

This report records the engineering work completed during the first two development phases. The focus was on defining the system architecture, designing the mechanical enclosures, assembling the board and wrist units, establishing the ESP32 toolchain, and bringing the BNO055 IMU and NEO 8M GPS into communication.

2. System Architecture Established

The system was divided into two embedded units. The board unit is mounted under the skateboard deck and is responsible for sensing, motion processing, GPS tracking, storage and later wireless telemetry. The wrist unit is carried by the rider and provides the OLED status display and haptic feedback. Both units use Seeed XIAO ESP32 C3 controllers.

The board does the work; the phone and the wrist only show it

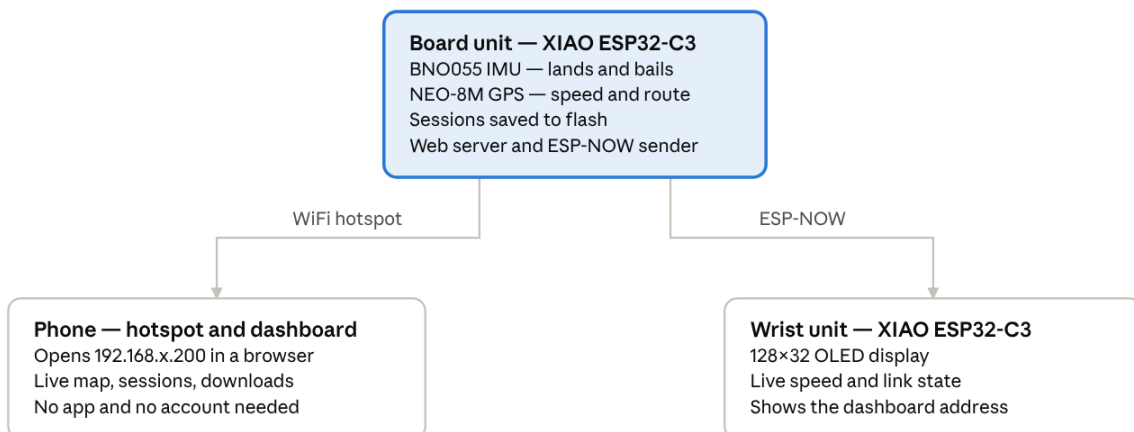


Figure 1. SkateGuard system architecture: board unit, phone dashboard and wrist unit.

2.1 Core Hardware and Interfaces

Parameter	Value	Purpose / implementation
Controller	Seeed XIAO ESP32 C3	Main controller for board and wrist units
IMU	Adafruit BNO055, I2C address 0x28	Orientation, Euler angles and gyro vector
IMU I2C	SDA GPIO6, SCL GPIO7	Board unit BNO055 interface
GPS	NEO 8M	Position, satellite count and speed
GPS UART	RX GPIO20, TX GPIO21, 9600 baud	Board GPS interface
Board battery ADC	GPIO4, 30k / 10k divider	Battery voltage measurement
Wrist display	SSD1306 128 x 32 OLED	Live rider information
Wrist motor	GPIO3	Haptic alert output

Parameter	Value	Purpose / implementation
Wrist status LED	GPIO8	Visual alert/status output
Battery	1500 mAh lithium cell	Portable power for each unit

3. Week 1: Foundation and Mechanical Design

The initial architecture was converted into a practical mechanical layout. The board enclosure was designed around the truck baseplate so the electronics remain attached to the deck without requiring a separate external box. The IMU was placed in a dedicated pod at the deck edge so its motion corresponds closely to deck rotation. The GPS antenna was positioned with an upward facing view, while the controller, battery and charging hardware were placed in the remaining enclosure volume.

- Both board and wrist housings were designed in CAD before printing.
- The wrist enclosure includes a display window, strap mounting points, indicator openings and a recessed USB C charging access point.
- The board enclosure provides mounting around the truck baseplate and separates the IMU, GPS and controller hardware.
- Component clearance and cable access were considered before fabrication.

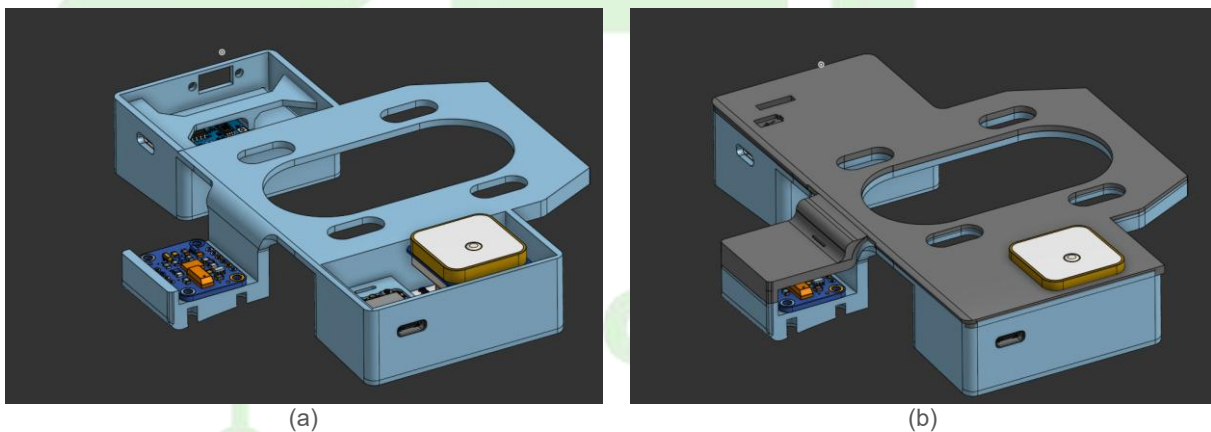


Figure 2. Board enclosure CAD around the truck baseplate: (a) internal layout with IMU pod and GPS, (b) closed assembly.

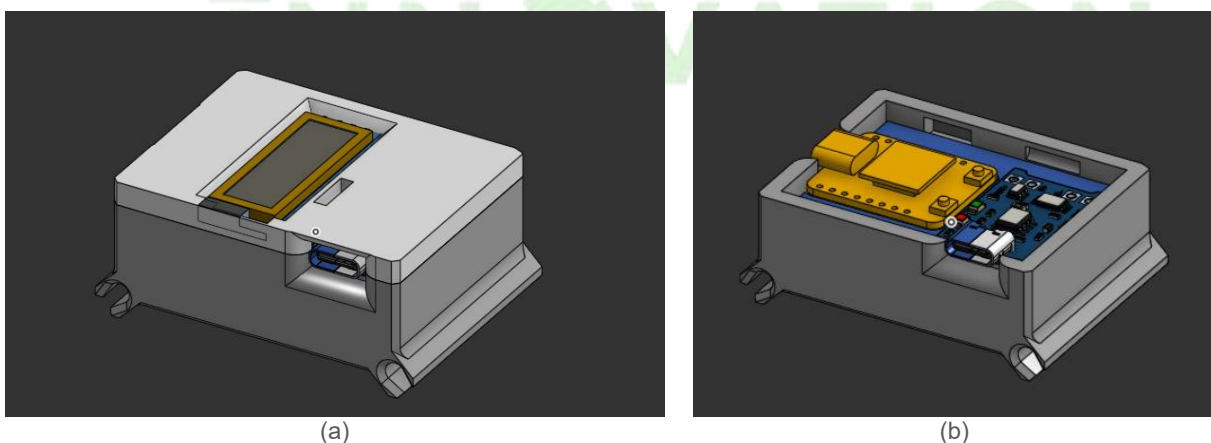


Figure 3. Wrist enclosure CAD: (a) OLED display window and strap lugs, (b) controller and charging module placement.

4. Detection Architecture Decisions

Two important changes were made during the foundation stage. The original MPU6050 plan was replaced with the BNO055. The BNO055 performs sensor fusion internally and provides orientation information directly, reducing the amount of fusion processing required in the ESP32 control loop. The original wobble concept was also changed from a fixed 25 km/h trigger to an oscillation based approach because GPS speed requires a valid satellite solution and can lag actual rider motion.

4.1 Mechanical and Detection Design Parameters

Parameter	Value	Purpose / implementation
Kickflip axis	GYRO_Y in current firmware	Current mounting assumption; real kickflip data is still required to confirm axis
Wobble speed gate	None in final detector	Oscillation pattern is used as the detection signature
Wobble threshold	3 oscillations	Number of counted oscillations required before wobble is reported
IMU fusion	BNO055 fused mode	Provides orientation and gyro data to controller
Board mounting	Truck baseplate	Keeps sensing hardware mechanically coupled to deck motion
Wrist mounting	Four corner strap lugs	Allows wearable operation while riding

5. Week 2: Electronics Assembly and Bring Up

The CAD designs were fabricated and both units were populated. The Arduino IDE toolchain was configured for the XIAO ESP32 C3 using ESP32 core 3.2.0. The board firmware established BNO055 communication over I2C and NEO 8M communication over UART1 at 9600 baud. Battery sensing was also integrated using the resistor divider arrangement.

The BNO055 was configured at I2C address 0x28. The firmware scans the I2C bus during initialization, providing an immediate indication of whether the sensor is electrically present. Battery initialization also records a raw ADC value so wiring errors can be detected during boot.

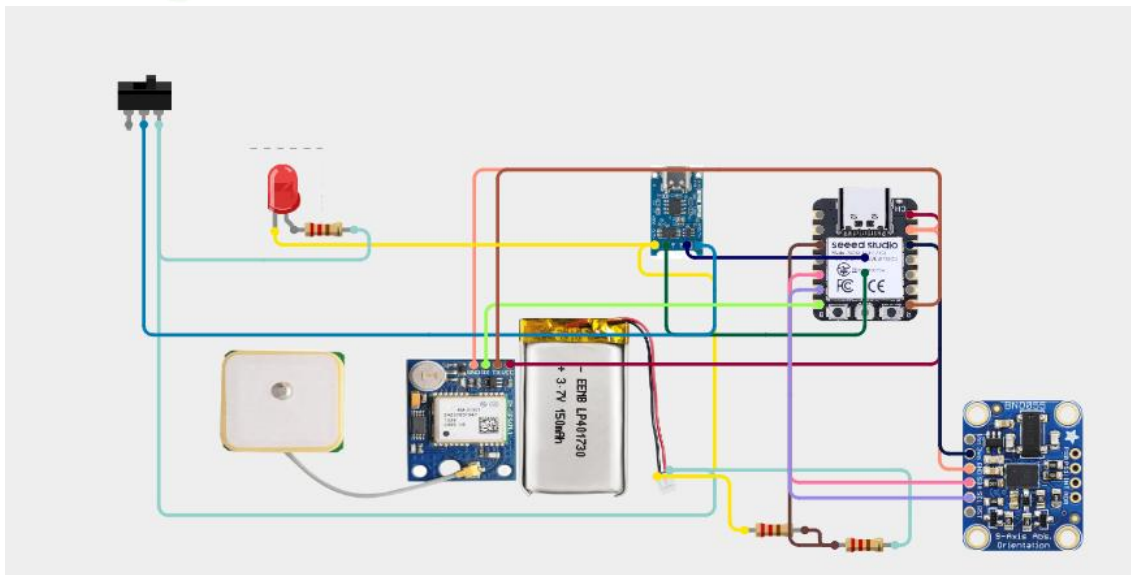


Figure 4. Board unit circuit diagram.

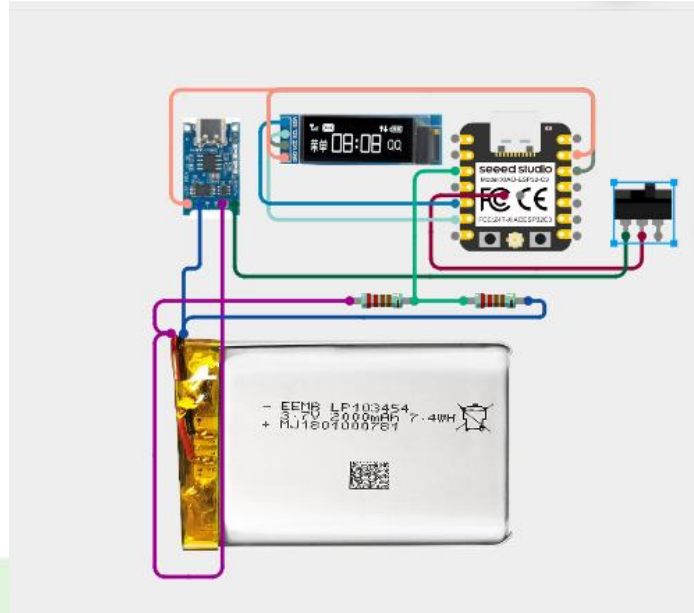


Figure 5. Wrist unit circuit diagram.

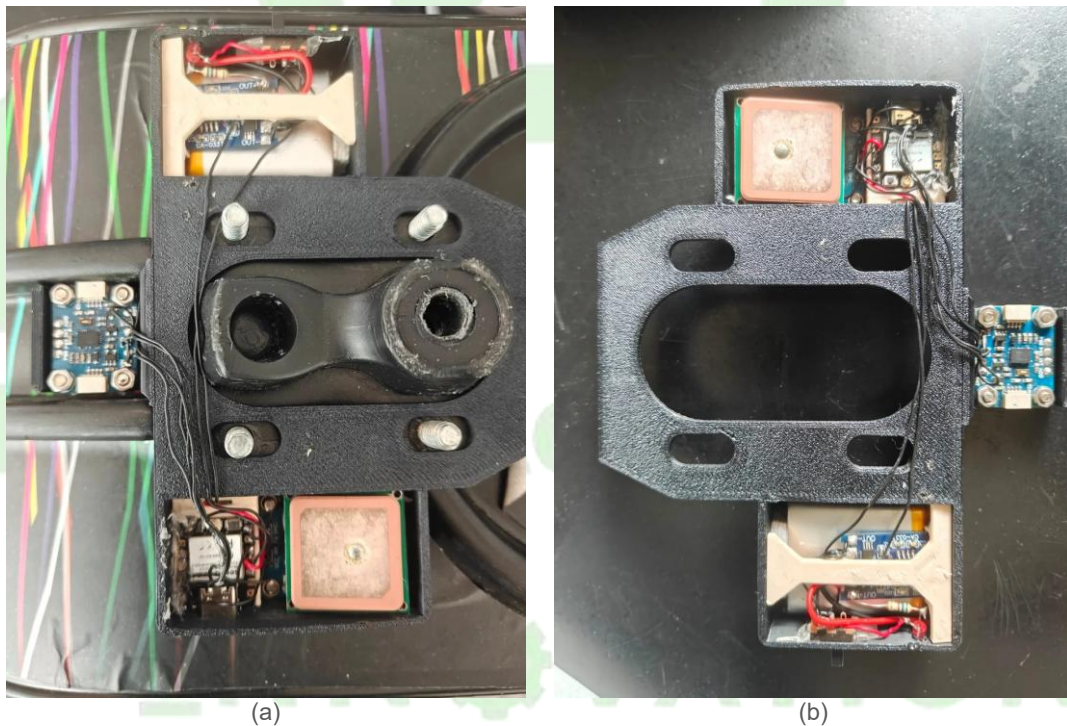


Figure 6. Assembled board unit mounted around the truck baseplate: (a) installed view, (b) top view of the enclosure.

5.1 Firmware and Electrical Parameters

Parameter	Value	Purpose / implementation
BNO055 address	0x28	I2C sensor address
I2C SDA	GPIO6 / D4	BNO055 data
I2C SCL	GPIO7 / D5	BNO055 clock
GPS baud rate	9600	NEO 8M UART communication
Battery divider	30k / 10k	Scales cell voltage for ADC
Battery ADC	GPIO4	Board battery measurement
ADC reference	3.3 V	Firmware ADC conversion reference

Parameter	Value	Purpose / implementation
Battery calibration	0.8815	Firmware voltage calibration factor
Board program partition	91% used	Remaining flash headroom became a later limitation
RAM usage	12%	Firmware memory usage reported in project report

6. Engineering Challenges Encountered

- Mechanical packaging required the GPS antenna, IMU, controller, battery and charging module to fit without blocking mounting or charging access.
- The correct IMU rotation axis could not be permanently assumed from the CAD orientation alone. The later firmware therefore logs gyro X and Z alongside gyro Y so the axis can be confirmed from a real kickflip.
- The original wobble speed threshold was not retained as the detection gate because GPS validity and update behavior make speed a poor direct trigger for the motion symptom.
- Final print material, layer height, infill and finished enclosure dimensions were not yet recorded and remain documentation items.

7. Sensor Validation Dataset

7.1 IMU Stabilization Test Data

Sample	Timestamp ms	Gyro y raw dps	Gyro y filtered dps	Tilt angle z deg	Filtered tilt angle z deg	Stability status
1	0	23.810	37.030	3.670	3.670	STABILIZING
2	100	37.030	38.670	4.180	3.780	STABILIZING
3	200	43.530	37.030	4.260	3.870	STABILIZING
4	300	40.300	37.030	4.090	3.920	STABILIZING
5	400	23.300	23.300	4.390	4.010	STABILIZING
6	500	7.740	7.740	4.640	4.140	STABILIZING
7	600	-1.890	-1.890	5.270	4.360	STABILIZING
8	700	-13.360	-13.360	4.430	4.380	STABILIZING
9	800	-19.710	-13.360	4.370	4.380	STABILIZING
10	900	-14.740	-13.360	4.090	4.320	STABILIZING

Table shows the first 10 rows from the test dataset for traceability. The complete CSV is retained separately.

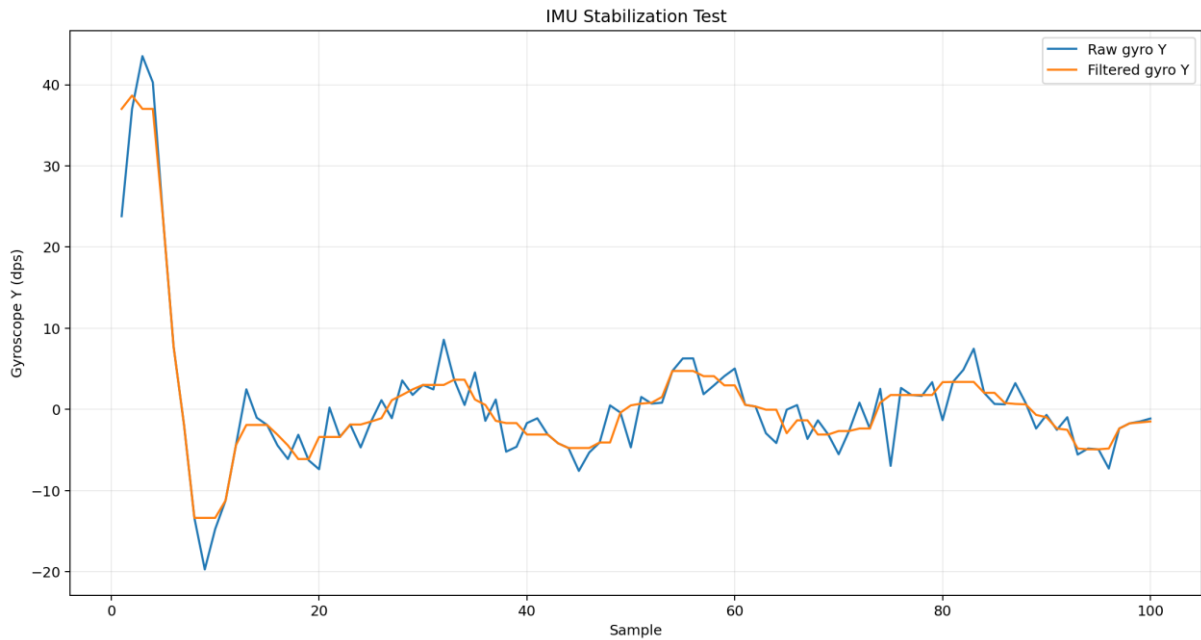


Figure 7. IMU raw and filtered gyro response used to visualize stabilization behavior.

8. Week 2 Completion Status

At the end of Week 2, the physical platform, embedded controllers and primary sensors were assembled and communicating. The project was therefore ready to move from hardware bring up into motion detection firmware and wireless communication.